

Human-Centered Design of Ethical Generative AI Interfaces for Healthcare

Abstract

Generative AI is reshaping healthcare from assisting in diagnostics to offering mental health support. While these technologies offer great promise, they also raise serious concerns around user trust, transparency, privacy, and equity. This thesis investigates how Human-Computer Interaction (HCI) principles can be integrated into the design of generative AI systems to enhance their ethical alignment, usability, and trustworthiness in patient-facing healthcare contexts. Through a combination of prototype development, usability testing, and ethical evaluation, this research explores how design choices impact user experience and perception of generative AI outputs. By focusing on real-world use cases—such as symptom checkers or mental health chatbots—this work develops a set of design guidelines for creating AI-powered healthcare tools that are transparent, respectful, and inclusive. The goal is to bridge the gap between technical AI performance and meaningful, human-centered interaction in sensitive health environments.

Thesis Overview

This thesis takes a multidisciplinary approach by combining HCI methodologies with AI ethics and design research. It begins by reviewing existing generative AI applications in healthcare and identifying common ethical and usability concerns. A functional prototype of a healthcare chatbot or advisory interface powered by a large language model (LLM) will then be developed. This system will be evaluated using HCI techniques such as heuristic evaluation, user testing, and semi-structured interviews.

The research will focus on how users perceive AI advice, understand system limitations, and make decisions based on AI-generated outputs. The study will also assess how different interface designs (e.g., disclaimers, confidence levels, interactive feedback) influence user trust and perceived reliability. The outcome will be a set of user-centered, ethically-informed design guidelines aimed at supporting safe and responsible generative AI integration in healthcare tools.

Objectives of the Thesis

1. To explore ethical and usability challenges in current generative AI applications within healthcare, especially those related to trust, transparency, and privacy.
2. To design and implement a functional prototype of a patient-facing generative AI tool (e.g., chatbot, advisor) using HCI best practices.
3. To evaluate user interaction with generative AI outputs through usability testing, heuristic evaluation, and qualitative feedback, focusing on patient comprehension, emotional responses, and decision-making.
4. To analyze user perception of ethical risks, including bias, misinformation, and unclear responsibility, and how these risks affect trust in the AI system.
5. To propose a set of ethical HCI design guidelines that promote inclusive, transparent, and responsible use of generative AI in healthcare settings.

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